

Smaran Rangarajan Bharadwaj

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EDUCATION

- **RV University, Bengaluru** · B.Tech (Hons.), Computer Science and Engineering Graduating: Aug 2026 · CGPA: 9.80
- **Sri Kumaran Children's Home, Bengaluru** · Class 12 (CBSE) - PCMC 2022 · Score: 95.2%

TECHNICAL SKILLS

Languages: Python3, Go, C, C++, Java, JavaScript, YAML
Cloud & DevOps: AWS, Docker, Kubernetes, Git, GCP, Jira, Confluence
Databases: ClickHouse, PostgreSQL, MySQL, MongoDB, SQLite
Frameworks: PyTorch, TensorFlow, Node.js, Flask, Streamlit, gRPC, RESTful APIs, Ollama, Kafka, Flink, Unsloth, LangChain
Relevant Coursework: AWS Networking Masterclass, Big Data Computing, Distributed Systems

EXPERIENCE

Software Development Intern Jan 2026 - Jul 2026
Aviatrix Systems Pvt. Ltd. Bengaluru

- Automated the CI/CD deployment pipeline with GitHub Actions, reducing release time for multi-hour builds and deployments by 68%.
- Architected and shipped an LLM-powered distributed multi-cloud (AWS/GCP) risk assessment ETL pipeline leveraging Flink and ClickHouse analyzing 22 resource types in under 100 seconds.
- Designed and shipped the scalable, cloud-native Data-Bridge backend in Go using gRPC and ClickHouse materialized views, enabling sub-second queries across 10M+ records.

ML Project Intern Aug 2025 - Jan 2026
Aeronautical Development Establishment (DRDO) Bengaluru

- Architected a multi-stage mission debriefing pipeline processing 10,000+ flight parameters across 6 missions in under 15 minutes with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems logic to classify overlapping continuous and discrete parameter anomalies.

ML Engineer Intern Jun 2025 - Aug 2025
GovernAIce Berkeley (Remote)

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a scalable batch document ingestion pipeline processing 12,000+ legal records across 6 languages with XLM-RoBERTa vector embedding generation.

LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Awarded merit scholarships for 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Heavens Hackathon for developing a distributed computing solution for refugee support systems.
- Co-designed and published a patent for Gelos, a fitness-focused interactive game, alongside Dr. Phani Kumar Pullela.

PROJECTS

MITM Attack Simulator Network Security, Flask, PostgreSQL

- Engineered a full-stack network security toolkit processing 15,000+ packets/sec, utilizing Scapy, raw socket programming, and Linux bridged networking to simulate MITM attacks and inspect IP traffic.

ESP32 Flow Meter Fault Detection FreeRTOS, Mosquitto (MQTT), TensorFlow Lite

- Engineered an IoT fault diagnosis system for ultrasonic flow meters on ESP32 running FreeRTOS, classifying system faults into 4 states with 91.67% accuracy.
- Deployed the optimized model via TensorFlow Lite and EloquentTinyML, achieving 300–400ms inference latency.

Obstacle Avoidance for Visually Impaired Computer Vision, YOLO, Transfer Learning

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP on a self collected dataset with 132ms average inference latency per frame.